

Anamul Hoque Emtiaj

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RESEARCH INTERESTS

Natural Language Processing, Software Engineering, Interdisciplinary AI Systems, and Applied Large Language Models.

EDUCATION

Bangladesh University of Engineering and Technology (BUET)

Feb 2020 – Mar 2025

B.Sc. in Computer Science and Engineering; CGPA: 3.79/4.00

Dhaka, Bangladesh

PUBLICATIONS

My name is in **bold**; * denotes equal contribution.

[C1] **Anamul Hoque Emtiaj**, Anindya Iqbal. “Auditing Bank Statement PDFs: A Multi-Tier Parser and Design Choices for the NL-to-SQL Query Layer Above It.” *11th Workshop on Financial Technology and Natural Language Processing (FinNLP 2026)*, co-located with EMNLP 2026, Budapest, Hungary. **Accepted**. [Code]

[J1] **Anamul Hoque Emtiaj***, Rakibul Hasan Rafi*, Muhammad Ali Nayeem, M. Sohel Rahman. “CRISMER: A Transformer-based Interpretable Deep Learning Approach for Genome-wide CRISPR-Cas9 Off-Target Prediction and Optimization.” *Array* (Elsevier). **Under revision**; revision submitted Jul 2026. [bioRxiv] [Code]

RESEARCH EXPERIENCE

Research Assistant, BUET CSE Incubator

Apr 2026 – Present

Supervisor: [Prof. Anindya Iqbal](#) | Interdisciplinary AI Systems

- **Auditing bank statements in natural language [C1]**: built a multi-tier parser (deterministic PyMuPDF for tabular pages, LlamaParse for the rest) that heuristically repairs parsing errors and standardizes statements into PostgreSQL, then audits them in natural language with a SQL agent. Compared zero-context, RAG and execution-feedback strategies for the agent, showing how far newer model generations reduce the need for a complex architecture.
- **Building-code compliance checking** (Ongoing): a code-generation agent that produces reusable and one-time checkers for building-code clauses against building models. Experiments under way toward an LLM4Code submission.

Research Intern, UIUC++ SRSE 2026

Jun 2026 – Present

Supervisors: [Prof. Darko Marinov](#) (UIUC), [Prof. Wenxi Wang](#) (UVA) | Summer Research in Software Engineering

- Working on formal verification of Rust programs using Verus: specification testing for LeetCode and Codeforces problems, and test-case generation. Selected to continue beyond August.

Undergraduate Thesis, Department of CSE, BUET

Dec 2023 – Mar 2025

Supervisor: [Prof. M. Sohel Rahman](#) | Computational Biology

- **CRISPR-Cas9 off-target prediction [J1]**: Designed CRISMER to predict off-target sites from DNA-guide-RNA sequence pairs: a multi-branch CNN extracting k -mer features, with transformer blocks for long-range dependencies across the encoded pair. Applied DeepSHAP and Integrated Gradients to attribute predictions to mismatch positions, and used the model to drive guide-RNA optimization.
- Achieved a 2–8% improvement in PR-AUC and F1 over recent state-of-the-art baselines on genome-wide off-target datasets. Supported by the RISE BUET Students’ Research Grant.

INDUSTRY EXPERIENCE

Oplusdojo | Dhaka, Bangladesh
Software Engineer — contracted to two North American startups

May 2025 – Present

Bot Food Corporation — Toronto, Canada

May 2026 – Present

- Full-stack engineer on a personal context management platform: users upload files and folders, connect online sources, and connect AI apps such as Claude and ChatGPT over MCP, so the retained context can be reused for personalized answers.
- Work across every end of the product — desktop app, web frontend, backend, AWS infrastructure, and monitoring with Sentry and PagerDuty.

EloraHQ — Seattle, WA, USA

May 2025 – Apr 2026

- Contributed to an LLM-powered real-time assistance platform for social care providers: voice-to-text transcription (Vosk, AWS Transcribe), sentiment analysis, and vector similarity search (Pinecone).
- Built a secure chatbot with guardrails and PII protection, and integrated voice call capabilities via AWS Connect within a Next.js, FastAPI and AWS architecture.

AWARDS & ACHIEVEMENTS

Research Grant: RISE BUET Students’ Research Grant, 2024 — awarded to our undergraduate thesis group.
Hult Prize: Hult Prize 2023 On-Campus Champion and Regional Round Participant (Milan Summit).

SKILLS

Languages: Python, C, C++, Java, Rust, JavaScript, SQL, Bash
ML & LLM: PyTorch, TensorFlow, scikit-learn, LangChain, LangGraph
Web & Data: FastAPI, Django, Django REST Framework, Next.js, Celery, PostgreSQL
Tools & Clouds: Git, GitHub, Docker, Kubernetes, AWS, Azure, DigitalOcean

TEST SCORES

TOEFL iBT (Sep 2026): overall **5.0/6.0**, with 5.0 in each of reading, listening, speaking and writing (CEFR C1).

REFERENCES

Dr. Anindya Iqbal: Professor, Department of CSE, BUET | anindya@cse.buet.ac.bd | *Research supervisor*
Dr. M. Sohel Rahman: Professor, Department of CSE, BUET | msrahman@cse.buet.ac.bd | *Thesis supervisor*